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Mukesh Patel School of Technology Management & Engineering

Log Book for capstone project

Department: Computer Science

Program: Bachelor of Technology

Semester: VII

Project Title: Agentic AI Framework for Context-Aware Semantic Understanding and Intelligent Assistance of Dynamic Web Forms

Name of Faculty Mentor: Dr. Nitin S. Choubey

Student Details

Std No.	Roll No	Name	Contact *(per. email)*
Student 1	E246	Ashmit Patil	edu.ashmit@gmail.com
Student 2	E265	Sanket Mali	sanketmali0909@gmail.com
Student 3	E247	Pinak Salunke	pinaksalunke@gmail.com

Group Members

Roll No.	Name	Sign (Students)	Grades (By Faculty Mentor)
E246	Ashmit Patil		
E265	Sanket Mali		
E247	Pinak Salunke		

Project Log Entries

Project Title: *Agentic AI Framework for Context-Aware Semantic Understanding and Intelligent Assistance of Dynamic Web Forms*

Week No.	Duration	Work Carried Out	Remarks by Faculty Mentor	Sign
1	13 Jul 2026 *to* 19 Jul 2026	• Formed the project group and met the faculty mentor to discuss possible project domains. • Shortlisted candidate problem statements in applied AI and web automation.		
2	20 Jul 2026 *to* 26 Jul 2026	• Identified repetitive manual data entry in online forms (admissions, scholarships, examinations, banking, government services) as the problem area. • Finalised the project title, aim, objectives and initial scope with the mentor.		
3	27 Jul 2026 *to* 02 Aug 2026	• Literature survey on LLM-based web agents and web-agent benchmarks. • Prepared summary notes of the surveyed papers.		
4	03 Aug 2026 *to* 09 Aug 2026	• Studied work on agent memory, knowledge representation, form understanding and form filling. • Reviewed research on form security and privacy risks of automated filling.		
5	10 Aug 2026 *to* 16 Aug 2026	• Surveyed supporting techniques: OCR and document information extraction, retrieval-augmented generation (RAG) and prompt injection. • Compared existing autofill approaches (browser autofill, password managers, form-filling agents) and identified the research gap.		
6	17 Aug 2026 *to* 23 Aug 2026	• Wrote the problem definition and listed the limitations of the existing system. • Defined functional and non-functional requirements, assumptions and constraints.		
7	24 Aug 2026 *to* 30 Aug 2026	• Selected the technology stack: FastAPI, React + Vite, MongoDB, LangGraph, OCR engines and a Chrome extension. • Tried candidate OCR engines and LLM providers on small sample inputs to confirm feasibility.		

Week No.	Duration	Work Carried Out	Remarks by Faculty Mentor	Sign
8	31 Aug 2026 *to* 06 Sep 2026	• Designed the system architecture, data flow and MongoDB database schema. • Designed the form-filling pipeline, decision rules and a shared ontology of personal fields. • Planned the security design: review-before-fill and keeping sensitive values away from the LLM.		
9	07 Sep 2026 *to* 13 Sep 2026	• Set up the project repository with backend, frontend and browser-extension modules. • Initialised the React + Vite frontend and the FastAPI backend with document upload and storage services (local and Cloudinary). • Implemented the first AI extraction agents and the document-processing pipeline.		
10	14 Sep 2026 *to* 20 Sep 2026	• Tested OCR and field extraction on sample documents and recorded failure cases. • Fixed bugs in the upload and extraction flow. • Refined the canonical field keys used to map document values to form fields.		
11	21 Sep 2026 *to* 27 Sep 2026	• Designed the privacy-first filling flow and a personal RAG store over the user's own documents. • Ran embedding and retrieval experiments for answering form questions from stored facts.		
12	28 Sep 2026 *to* 04 Oct 2026	• Integrated privacy-first form filling and the personal RAG module into the backend. • Connected the frontend and extension to the new filling endpoints.		
13	05 Oct 2026 *to* 11 Oct 2026	• Improved form-field extraction metadata in the extension (labels, names, types and options). • Fixed handling of grouped radio buttons. • Began a full audit of the codebase.		

Week No.	Duration	Work Carried Out	Remarks by Faculty Mentor	Sign
14	12 Oct 2026 *to* 18 Oct 2026	• Wrote the project audit, implementation plan and testing strategy. • Hardened authentication, configuration, data integrity and deletion cascades. • Added fact provenance, grounded RAG and a single batch LangGraph pipeline for semantic form filling. • Built the Chrome extension review-before-fill flow and the dashboard (review, conflicts, document Q&A); integrated PaddleOCR. • Added unit, integration and browser end-to-end tests, a prompt-injection filter and secret redaction; ran the evaluation and OCR benchmarks. • Prepared the synthetic demo dataset, the demonstration guide and the first draft of the project report.		
15	19 Oct 2026 *to* 25 Oct 2026	• Finalised the project report (Black Book): chapters, figures, results and testing. • Ran the plagiarism and AI-detection checks and made the required corrections. • Prepared the final print copy, final submission documents and this log book.		

Date: _____

Place: Shirpur

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